ANDREW FISCHER

Victoria BC, Canada | (778) 676-1477 | andrewfischer@uvic.ca | Andrew-Fischer | andrewbfischer.com

Summary

Ambitious, extremely hardworking 4th-year electrical and computer engineering student with hands-on experience developing complete embedded systems. I have over 3 years of documented programming experience through work, clubs, and personal projects. In my most recent internship, I developed an annunciator panel used for firefighter training, which strengthened my skills in robust embedded system design and long-term maintainability.

Skills

Languages & Protocols: C, C++, Python, MATLAB, Lua, VHDL, CMake, Git
Embedded Systems: Architectures: AVR, ARM (ARMv6-M), Interfaces: CAN, UART, SPI, I2C, RTOS: FreeRTOS
Design Software: KiCAD, Altium, AutoCAD, LTspice
Lab Equipment: Logic Analyzer, Oscilloscope, Digital Multimeter, Signal Generator
Qualifications: Licensed Amateur Radio Operator - Basic and Advanced (Honours)

Technical Projects

Supercapacitor Bike

- Designed a complete embedded system to drive a modified BLDC motor.
- Developed lightweight, high-performance ATmega328P firmware, including I2C, SPI, EEPROM, Timer & PWM drivers using AVR-C and AVR-ASM.
- Designed a dynamic ring buffer to reduce memory overhead and manage bus-busy I2C requests.
- See project here.

Work Experience

Electrical Engineer (Internship) — JAN. 2025 – AUG. 2025 (8 MO)
ContainerWest — Richmond, BC

- Independently designed (KiCAD, C++), tested, documented and presented a working TRL-6 prototype of an expandable annunciator panel capable of monitoring rooms for smoke and fire, allowing firefighters to interact with the panel to simulate events.
- Developed ladder logic programs for OMRON NX1P2 PLCs to automate tasks and monitor live fire burn props to ensure safe operation.
- Gained practical hands-on experience with high-power equipment, including 3-phase AC induction motors, incinomites, and transformers.
- Collaborated with tradespeople and electricians, providing technical guidance and support for installations and troubleshooting.
- Developed ENEFEN-approved system flow graphs and electrical schematics in AutoCAD and performed the associated on-site installations and control system modifications.
- Performed independent site visits and troubleshooting, working directly with clients to ensure reliable and safe operation of the live fire burn props.

Education

University of Victoria — Bachelor of Electrical Engineering, 4th year (CGPA : 3.7/4.0)